

1. CREATE BOOKS TABLE

1 • DROP DATABASE IF EXISTS bookflow_db;
 2 • CREATE DATABASE bookflow_db;
 3 • USE bookflow_db;
 4
 5 • CREATE TABLE Books
 6 (
 7 book_id INT AUTO_INCREMENT PRIMARY KEY,
 8 title VARCHAR(200) NOT NULL,
 9 author VARCHAR(150) NOT NULL,
 10 genre VARCHAR(100),
 11 description TEXT,
 12 published_year INT
 13);
 14
 15
 16

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 2	14:32:08	DROP DATABASE IF EXISTS bookflow_db	2 row(s) affected	0.125 sec
✓ 3	14:32:29	CREATE DATABASE bookflow_db	1 row(s) affected	0.031 sec
✓ 4	14:32:33	USE bookflow_db	0 row(s) affected	0.000 sec
✓ 5	14:34:07	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected	0.062 sec

15 • DESCRIBE Books;
 16
 17
 18

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(200)	NO		NULL	
	author	varchar(150)	NO		NULL	
	genre	varchar(100)	YES		NULL	
	description	text	YES		NULL	
	published_year	int	YES		NULL	

Result 1 × Read Only Cont

Output

Action Output

#	Time	Action	Message
✓ 3	14:32:29	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 4	14:32:33	USE bookflow_db	0 row(s) affected
✓ 5	14:34:07	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected
✓ 6	14:39:04	DESCRIBE Books	6 row(s) returned

2. INSERT BOOK DATA

```
17 • INSERT INTO Books
18 (
19     title,
20     author,
21     genre,
22     description,
23     published_year
24 )
25 VALUES
26
27 (
28     'The Martian',
29     'Andy Weir',
30     'Science Fiction',
31     'An astronaut struggles to survive alone on Mars.',
32     2011
33 ),
34
35 (
36     'Dune',
37     'Frank Herbert',
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 4	14:32:33	USE bookflow_db	0 row(s) affected	0.000 sec
✓ 5	14:34:07	CREATE TABLE Books (book_id INT AUTO_INCREMENT ...	0 row(s) affected	0.062 sec
✓ 6	14:39:04	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
✓ 7	14:42:05	INSERT INTO Books (title, author, genre,....	10 row(s) affected Records: 10 Duplicates: 0 W...	0.016 sec

3. DISPLAY BOOKS

```
107 • SELECT
108     book_id,
109     title,
110     author,
111     genre,
112     published_year
113 FROM Books
```

manually get help for the current caret position or to toggle automatic help.

Result Grid

	book_id	title	author	genre	published_year
▶ 1		The Martian	Andy Weir	Science Fiction	2011
2		Dune	Frank Herbert	Science Fiction	1965
3		The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4		Pride and Prejudice	Jane Austen	Romance	1813
5		The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6		Foundation	Isaac Asimov	Science Fiction	1951
7		The Hobbit	J.R.R. Tolkien	Fantasy	1937
8		Enders Game	Orson Scott Card	Science Fiction	1985
9		Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10		Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
• NULL		NULL	NULL	NULL	NULL

Books 2 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 5	14:34:07	CREATE TABLE Books (book_id INT AUTO_INCREMENT ...	0 row(s) affected	0.062 sec
✓ 6	14:39:04	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
✓ 7	14:42:05	INSERT INTO Books (title, author, genre, description...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
✓ 8	14:44:06	SELECT book_id, title, author, genre, published_y...	10 row(s) returned	0.000 sec / 0.000 sec

4. ADD EMBEDDING COLUMN

```
116 • ALTER TABLE Books
117   ADD COLUMN embedding JSON;
118
119 • DESCRIBE Books;
120
121
122
123
```

disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Result 3 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 9	14:46:43	ALTER TABLE Books	0 row(s) affected	0.015 sec
✗ 10	14:46:49	ADD COLUMN embedding JSON	Error Code: 1064. You have an error in your SQL syntax; check...	0.000 sec
✓ 11	14:49:13	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
✓ 12	14:49:41	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

5. INSERT MOCK VECTORS

1. BOOK 1 - MARTIN

```
121 • UPDATE Books
122   SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
123   WHERE book_id = 1;
124
125
126
127
```

Context Help Snip

Output

Action Output

#	Time	Action	Message	
✗ 10	14:46:49	ADD COLUMN embedding JSON	Error Code: 1064. You have an error in your SQL syntax; check...	0.
✓ 11	14:49:13	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.
✓ 12	14:49:41	DESCRIBE Books	7 row(s) returned	0.
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.

2. DUNE

```
125 • UPDATE Books
126     SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
127     WHERE book_id = 2;
128
129
130
```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 11	14:49:13	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 12	14:49:41	DESCRIBE Books	7 row(s) returned
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

3. HITCHHIKERS GUIDE

```
129 • UPDATE Books
130     SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
131     WHERE book_id = 3;
132
133
```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 12	14:49:41	DESCRIBE Books	7 row(s) returned
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

4. PRIDE AND PREJUDICE

```
133 • UPDATE Books
134     SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
135     WHERE book_id = 4;
136
137
138
```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	14:54:50	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

5. GREAT GATSBY

```
137 • UPDATE Books
138     SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
139     WHERE book_id = 5;
140
141
142
```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	14:54:50	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	14:55:40	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

6. FOUNDATION

```
141 • UPDATE Books
142     SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
143     WHERE book_id = 6;
144
145
146
```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	14:54:50	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	14:55:40	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	14:56:14	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

7. THE HOBBIT & ENDERS GAME

```
145 • UPDATE Books
146     SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
147     WHERE book_id = 7;
148
149 • UPDATE Books
150     SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
151     WHERE book_id = 8;
152
153
```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 17	14:55:40	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	14:56:14	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 19	14:57:09	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	14:57:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

8. HARRY POTTER & TWENTY THOUSAND LEAGUES

```

153 • UPDATE Books
154     SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
155     WHERE book_id = 9;
156
157 • UPDATE Books
158     SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
159     WHERE book_id = 10;
160
161
162

```

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 19	14:57:09	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	14:57:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 21	15:01:30	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

9. VERIFY ALL VECTORS

```

161 • SELECT
162     book_id,
163     title,
164     embedding
165 FROM Books
166 ORDER BY book_id;
167

```

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book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
NULL	NULL	NULL

Books 4 x

Apply

Revert

[Context Help](#)

Output

Action Output

#	Time	Action	Message
✓ 20	14:57:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 21	15:01:30	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 23	15:04:21	SELECT book_id, title, embedding FROM Books ORDE...	10 row(s) returned

10. L2 DISTANCE

168 • **SELECT**

169 book_id,

170 title,

171 author,

172 genre,

173 ROUND(
174 SQRT(

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: | Fetch rows: | **Result Grid** | Form Editor | Field Types

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Andy Weir	Science Fiction	0
	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
	2	Dune	Frank Herbert	Science Fiction	0.0866

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Result 5 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 21	15:01:30	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 23	15:04:21	SELECT book_id, title, embedding FROM Books ORDE...	10 row(s) returned
✓ 24	15:10:15	SELECT book_id, title, author, genre, ROUND(...	3 row(s) returned

11. COSINE SIMILARITY

215 • **SELECT**

216 book_id,

217 title,

218 author,

219 genre,

220 ROUND(
221 (
222 CAST(
223 JSON_EXTRACT(embedding, '\$[0]')

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: | Fetch rows: | **Result Grid** | Form Editor | Field Types

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

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Result 6 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 23	15:04:21	SELECT book_id, title, embedding FROM Books ORDE...	10 row(s) returned
✓ 24	15:10:15	SELECT book_id, title, author, genre, ROUND(...	3 row(s) returned
✓ 25	15:14:09	SELECT book_id, title, author, genre, ROUND(...	3 row(s) returned